# Section 2.6

## Elimination = Factorization
## = LU Decomposition

# Benefits of the matrixinverse.

- **The matrix inverse "stores" the result of Gaussian elimination, allowing solutions to be found with new right-hand sides easily and efficiently.**
  - **Gaussian elimination for a system of $n$ equations and $n$ unknowns requires about $\left(\frac{2}{3}\right)n^3$ total arithmetic operations; it is an $O\left(n^3\right)$ problem.**
  - **Matrix multiplication of an $n \times n$ matrix and an $n \times 1$ vector requires about $2n^2$ total arithmetic operations; it is an $O\left(n^2\right)$ problem.**
- **The matrix inverse provides a framework in matrix algebra which can be used to develop new approaches to solve linear system.**

# Limitations of the matrixinverse

- The matrix inverse does not exist for non-square matrices, nor for all square matrices.
- Systems of equations in fewer equations than unknowns may have a solution for all possible right-hand sides, just not one that is unique.
- For a system that has many solutions, the matrix inverse does not work.
- Systems of any size will have at least some right-hand sides that have a solution, although not necessarily unique. However, the matrix inverse could not determine which right-hand sides lead to a consistent system.
- **All these problems we were able to handle with Gaussian Elimination.**

# Goal of the LU decomposition

- **Store information about each step of forward elimination in Gaussian elimination. The forward elimination is the most computationally costly part of the problem.**
- **Similar reduction of work for multiple right-hand-sides as the matrix inverse.**
- **Applicable to both square and non-square systems. Used to derive for a unique solution or multiple solutions and determine whether a right-hand side leads to a consistent system.**

# A summary of LU decomposition matrices

**We will use the following notation in our discussion**

- $A$: **The original matrix (Any Size).**
  - **In general, this will be the coefficient matrix of a linear system, not the augmented matrix.**
- $U$: **The echelon (upper triangular) form of $A$.**
  - **This is the matrix found after only the forward elimination step.**
- $L$: **A lower triangular matrix.**
  - **This matrix stores the information about forward elimination.**
- $P$: **A permutation matrix (Section 2.7).**
  - **If row exchanges were used, this matrix stores them.**

# Two forms of LU decomposition.

- **When row exchanges are not needed, the LU decomposition has the form:**

$$A = LU$$

  - **The original matrix is expressed as a product of a lower triangular and an upper triangular matrix.**
- **When row exchanges are needed (Do this first!!!), the LU decomposition has the form (Section 2.7 Permutations):**

$$PA = LU$$

  - **The original matrix first has its rows rearranged, then the result is expressed as a product of a lower triangular and an upper triangular matrix.**

# Benefit of the LU decomposition

- **Many matrix decompositions are used in linear algebra and share the common feature of expressing a matrix as a product of two, but both have a special form:**

$$A\vec{x} = \vec{b}, \qquad A = LU \rightarrow LU\vec{x} = \vec{b} \rightarrow L\vec{c} = \vec{b}, \qquad U\vec{x} = \vec{c}$$

- **Step 1: Solve the system $L\vec{c} = \vec{b}$, to find $\vec{c}$. Easy: $L$ is lower triangular.**
- **Step 2: Solve the system $U\vec{x} = \vec{c}$, to find $\vec{b}$. Easy: $U$ is upper triangular.**

# Gaussian Elimination is reversible.

- **Every elementary row operation has a corresponding inverse operation that "undoes" the effect (Important in finding $L^{-1}$):**
  - **Addition of a scalar multiple is reversed by subtracting the same scalar multiple.**
  - **Scalar multiplication is reversed by multiplying by the inverse of the scalar.**
  - **Row exchange is reversed by repeating the same exchange.**

| Original Operation | Inverse Operation |
|---|---|
| $R_i + kR_j \rightarrow R_i$ | $R_i - kR_j \rightarrow R_i$ |
| $kR_i \rightarrow R_i$ | $\frac{1}{k}R_i \rightarrow R_i$ |
| $R_i \leftrightarrow R_j$ | $R_i \leftrightarrow R_j$ |

# Elementary matrices are non-singular.

- **Every elementary row operation has a corresponding inverse operation that "undoes" the effect:**

$$A \xrightarrow{R_f} A' \xrightarrow{R_r} A$$

- **We can define matrices for the forward and inverse operations:**

$$A' = E_f A, \qquad A = E_r A'$$

- **The elementary matrix for the inverse operation is the inverse of the elementary matrix for the forward operation!**

$$A = E_r E_f A, \forall A \rightarrow E_r E_f = I$$
$$A' = E_f E_r A', \forall A \rightarrow E_f E_r = I$$
$$E_r = (E_f)^{-1}$$

# Example: Inverse of an elimination matrix

- Given E is an elementary matrix that subtracts 5 times row 1 from row 2, find the inverse $E^{-1}$?

Answer: According to the rule, the inverse operation will reverse the subtraction by addition. $E^{-1}$ adds 5 times row 1 to row 2.

$$\begin{matrix} \boldsymbol{E} \textbf{ subtracts} \\ \boldsymbol{E^{-1}} \textbf{ adds} \end{matrix} \qquad E = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{-5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

# Example: inverse of an elimination matrix

- Given F subtracts 4 times row 2 from row 3, find $F^{-1}$?

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \quad \text{and} \quad F^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

# Interesting example

$$FE = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ \mathbf{20} & -4 & 1 \end{bmatrix} \quad \text{is inverted by} \quad E^{-1}F^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{5} & 1 & 0 \\ \mathbf{0} & \mathbf{4} & 1 \end{bmatrix}.$$

$E$ subtracts 5 times row 1 from row 2. Then $F$ subtracts 4 times the *new* row 2 (changed by row 1) from row 3. ***In this order $FE$, row 3 feels an effect from row 1.***

In the order $E^{-1}F^{-1}$, that effect does not happen. First $F^{-1}$ adds 4 times row 2 to row 3. After that, $E^{-1}$ adds 5 times row 1 to row 2. There is no 20, because row 3 doesn't change again. ***In this order $E^{-1}F^{-1}$, row 3 feels no effect from row 1.***

# Generalization: The forward elimination matrix.

- **Recall that forward elimination can be represented by a product of elementary matrices:**

$$A \xrightarrow{R_{f1}} A_{f1} \xrightarrow{R_{f2}} A_{f2} \xrightarrow{R_{f3}} \cdots \xrightarrow{R_{fr-1}} A_{fr-1} \xrightarrow{R_{fr}} A_{fr} = U$$

$$A \rightarrow E_{f1}A \rightarrow E_{f2}(E_{f1}A) \rightarrow \cdots \rightarrow E_{fr}\left(E_{fr-1}\left(\cdots E_{f2}(E_{f1}A)\right)\right) = U$$

$$FA = U, \qquad F = \left(\prod_{i=0}^{r-1} E_{fr-i}\right)$$

# Generalization: The reverse elimination matrix reverses forward elimination.

- **We can apply the same idea, in reverse, to obtain a matrix that reverses elimination:**

$$U = A_{fr} \xrightarrow{R_{r1}} A_{fr-1} \xrightarrow{R_{r2}} A_{fr-2} \xrightarrow{R_{r3}} \cdots \xrightarrow{R_{rr-1}} A_{f1} \xrightarrow{R_{rr}} A$$

$$U \rightarrow E_{r1}U \rightarrow E_{r2}(E_{r1}U) \rightarrow \cdots \rightarrow E_{rr}\left(E_{rr-1}\left(\cdots E_{r2}(E_{r1}U)\right)\right) = A$$

$$RU = A, \qquad R = \left(\prod_{i=0}^{r-1} E_{rr-i}\right)$$

# Summary: the forward elimination matrix and back substitution matrix.

$$FA = U, \qquad F = \left(\prod_{i=0}^{r-1} E_{fr-i}\right) \qquad\qquad RU = A, \qquad R = \left(\prod_{i=0}^{r-1} E_{rr-i}\right)$$

- **The elementary matrices for the forward and reverse steps are related: the (i+1)th reverse step is the inverse of the (r-i)th forward step (and vice versa). Thus:**

$$E_{ri+1} = (E_{fr-i})^{-1} \qquad\qquad E_{fi+1} = (E_{rr-i})^{-1}$$

$$R = \left(\prod_{i=0}^{r-1} E_{rr-i}\right) = \left(\prod_{i=1}^{r} (E_{fi})^{-1}\right) = \left(\prod_{i=0}^{r-1} E_{fr-i}\right)^{-1} = F^{-1}$$

# Summary: the forward elimination matrix and back substitution matrix.

- **The inverse elimination matrix is the product of the inverse of the forward elimination matrices, with the product order reversed:**

$$FA = U, \qquad F = \left(\prod_{i=0}^{r-1} E_{fr-i}\right) \qquad\qquad A = F^{-1}U, \qquad F^{-1} = \left(\prod_{i=1}^{r} (E_{fi})^{-1}\right)$$

- **What do $F$ and $F^{-1}$ look like, and can we compute them easily?**

# Generally, forward elimination only needs one or two types of row operation.

**When we do the forward elimination algorithm, two eliminations are needed.**

- **Primary elimination operation:** $R_k - \left({}^{a_{kj}}/_{a_{ij}}\right)R_i \to R_k$
- **Occasional row exchange:** $R_i \leftrightarrow R_j$

# Elementary elimination matrices.

- **An example:**

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 - 3R_1 \to R_3} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = E$$

- **General form:**

$R_k - \left({}^{a_{kp}}/_{a_{lp}}\right) R_l \to R_k$ ($l$ is the pivot row and $p$ is the pivot column)

$$E = [e_{ij}], e_{ij} = \begin{cases} 1, & if\ i = j \\ -\left({}^{a_{kp}}/_{a_{lp}}\right), & if\ i = k, j = l \\ 0, & otherwise \end{cases}$$

- **E is an identity matrix with one non-zero off-diagonal element.**
- **Scaling factor should be located at row to be replaced and column matching the row used for elimination – always below the diagonal for forward elimination ($k > l$).**

# The product of (forward) elimination matrices is lower triangular.

- **The elementary matrices of forward elimination are lower triangular, with ones on the diagonal.**
- **The product of any such matrices is lower triangular with ones on the diagonal.**
- **An example:** $A \xrightarrow{(1)\ R_2-2R_1\to R_2} \xrightarrow{(2)\ R_3-3R_1\to R_3} \xrightarrow{(3)\ R_3+R_2\to R_3} U$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$F = E_3E_2E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -5 & 1 & 1 \end{bmatrix}$$

# The product of the inverse elimination matrices is lower triangular.

- **The inverse of the forward elimination steps are just the same steps applied in reverse order with a sign change.**

**Forward:** $R_k - \left({}^{a_{kp}}/_{a_{lp}}\right)R_l \to R_k$ **Inverse:** $R_k + \left({}^{a_{kp}}/_{a_{lp}}\right)R \to R_k$

- **The same properties must thus apply.**
- **An example:** $U \xrightarrow{(-3)\ R_3 - R_2 \to R_3} \xrightarrow{(-2)\ R_3 + 3R_1 \to R_3} \xrightarrow{(-1)\ R_2 + 2R_1 \to R_2} A$

$$E_{-1} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} E_{-2} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ +3 & 0 & 1 \end{bmatrix}, E_{-3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$R = F^{-1} = E_{-1}E_{-2}E_{-3} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix}$$

# Forward elimination gives L and U without row changes.

- **We can write forward elimination and its inverse as a matrix product.**

$$FA = U, \qquad F = \left(\prod_{i=0}^{r-1} E_{fr-i}\right) \qquad\qquad A = F^{-1}U, \qquad F^{-1} = \left(\prod_{i=1}^{r} (E_{fi})^{-1}\right)$$

- **If no row exchanges are involved, $F$ and $F^{-1}$ are both lower triangular with ones on the diagonal.**
- $F^{-1}$ **is** $L$:

$$L \equiv F^{-1} \rightarrow A = LU, \qquad U = L^{-1}A$$

# Section 2.7

## Permutations of Matrix

# Permutation matrices: Elementary row exchange matrices.

- **An example**

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E$$

- **General form:** $R_k \leftrightarrow R_l$

$$E = [e_{ij}], e_{ij} = \begin{cases} 1, & if\ i = j, \quad i, j \neq k, l \\ 1, & if\ i = k, j = l \\ 1, & if\ i = l, j = k \\ 0, & otherwise \end{cases}$$

- **An identity matrix with two rows swapped.**

# Permutation matrices: The product of row exchange matrices

- **Each multiplication corresponds to a row swap. Then an identity matrix with scrambled rows follows!**
- **There will be exactly one "1" in every row and in every column, and all other entries will be "0".**
- **An example:**

$$A \xrightarrow{(1)\ R_1 \leftrightarrow R_3} \xrightarrow{(2)\ R_2 \leftrightarrow R_3} A'$$

$$E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$P = E_2 E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

# Isolating row exchanges lead to PA=LU.

- **If row changes are necessary, please switch the rows before starting elimination.**

$$A \xrightarrow{Row\ Exchanges} A' \xrightarrow{Elimination} U$$

$$A' = PA$$

$$U = L^{-1}A' = L^{-1}PA$$

$$PA = LU$$

- $P$ **is the matrix that executes row exchanges.**
- $L$ **is the inverse of the matrix that executes the elimination steps.**

# A simple way to generate L.

$$A \xrightarrow[A' = PA,]{Row\ Exchanges} A' \xrightarrow[U = L^{-1}A']{Elimination} U$$

$$PA = LU$$

- $L$ **is a lower triangular matrix that reverses the elimination steps, after we have appropriately arranged the rows:**

$$L \equiv F^{-1} = \left( \prod_{i=1}^{r} (E_{fi})^{-1} \right)$$

# A simple way to generate L.

$$L \equiv F^{-1} = \left( \prod_{i=1}^{r} (E_{fi})^{-1} \right)$$

- **Each matrix in the product is the inverse of the corresponding forward elimination matrix.**
- **The reverse step involves the same operation with a different sign:**

  **Forward:** $R_i - kR_j \rightarrow R_i$ **Reverse:** $R_i + kR_j \rightarrow R_i$

- **We can directly write down these matrices as we perform elimination!**

# A simple way to generate L.

- **We can directly write down these matrices as we perform elimination!**

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow[(2): R_3 + R_1 \to R_3]{(1): R_2 - 2R_1 \to R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{(3): R_3 + 3R_2 \to R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = U$$

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ +1 & 0 & 1 \end{bmatrix}, E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & +3 & 1 \end{bmatrix}$$

$$(E_1)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, (E_2)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}, (E_3)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

# A simple way to generate L.

$$L \equiv F^{-1} = \left( \prod_{i=1}^{r} (E_{fi})^{-1} \right)$$

**Note: The product has the individual scaling terms in the appropriate positions!**

$$L = (E_1)^{-1}(E_2)^{-1}(E_3)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ +2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}$$

# A simple way to generate L.

**Given an $m \times n$ coefficient matrix $A$:**

- **Create an $m \times m$ empty matrix that will become $L$.**
- **Place ones along the diagonal of $L$ and zeros above it.**
- **Apply Gaussian Elimination (forward steps) converting $A \to U$.**
- **With each elimination step, note the row of the pivot: call this $p$.**
- **Fill in column $p$ of $L$ (below the diagonal) with the multiple of the pivot row that was *subtracted* from each row below it:**

$$R_k - xR_p \to R_k \Rightarrow l_{kp} = x$$

- **Take care not to forget the cases where $x = 0$.**

# A simple way to generate L.

- **An example:**

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \xrightarrow[(2):R_3+R_1\to R_3]{(1):R_2-2R_1\to R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{(3):R_3+3R_2\to R_3} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix} = L$$

$$PA = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = LU$$

# Be careful with row exchanges!

- **Note: The direct approach to L works ONLY when there are no row exchanges applied during the elimination steps.**
- **If you do need row exchanges, they need to be applied first!**
- **"By-hand" approach (more work, but straightforward):**
  1. **Perform forward elimination to identify any necessary row exchanges.**
  2. **Apply these to the original matrix and repeat forward elimination to generate L.**
- **"Computational" approach (less work, but error-prone):**
  - **When we reach a row exchange, reconstruct the sequence of operations that would have been needed and adjust L directly.**

# LU decomposition checkpoints.

- The $U$ matrix is the end point of forward elimination.
- Checkpoints:
  - If $U$ has no zero rows, then the system has a solution for every possible right-hand side.
  - If $U$ has a pivot in every column (i.e. there are no free variables) then any solution must be unique.
  - For square matrices, these must both be true (for a non-singular system) or both be false (for a singular system).
  - For non-square matrices, these are independent checkpoints!

# Solving systems with LU decomposition.

- **The LU decomposition exists for EVERY matrix.**
- **It can be applied to ANY linear system.**

- **Consider:** $A\vec{x} = \vec{b}, \quad PA = LU$

$$LU\vec{x} = P\vec{b} \rightarrow \vec{b}' = P\vec{b}, \qquad L\vec{c} = \vec{b}', \qquad U\vec{x} = \vec{c}$$

1. **Re-order RHS, if needed:** $\vec{b}' = P\vec{b}$
2. **Solve Lower Triangular system:** $L\vec{c} = \vec{b}'$
3. **Solve Upper Triangular system:** $U\vec{x} = \vec{c}$

- **If $\vec{c}$ has a non-zero entry where $U$ has a zero row, no solution.**
- **If $U$ has a free variables, $\vec{x}$ is not unique**

# Solving systems with LU decomposition.

$$A\vec{x} = \vec{b}, PA = LU$$
$$\vec{b}' = P\vec{b}$$
$$L\vec{c} = \vec{b}'$$
$$U\vec{x} = \vec{c}$$

- **Example: Solve the system** $A\vec{x} = \vec{b}$

  **for** $\vec{b} = [1 \quad -1 \quad 2]^T$

  **given the LU decomposition of A,** $PA = LU$

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \; L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

# Solving systems with LU decomposition.

$$A\vec{x} = \vec{b}, PA = LU$$
$$\vec{b}' = P\vec{b}$$
$$L\vec{c} = \vec{b}'$$
$$U\vec{x} = \vec{c}$$

- **Step 1:** $\vec{b}' = P\vec{b} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$

- **Step 2:**

$$[L \quad \vec{b}'] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 1 \\ -1 & -3 & 1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & -3 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -8 \end{bmatrix} = [I \quad \vec{c}]$$

- **Step 3:**

$$[U \quad \vec{c}] = \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & -1 & -8 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & -6 \\ 0 & -1 & 0 & 5 \\ 0 & 0 & 1 & 8 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 8 \end{bmatrix} = [I \quad \vec{x}]$$

# Understanding LU decomposition

$$[A \quad \vec{b}] \xrightarrow{Row\ Exchanges} [A \quad \vec{b}'] \xrightarrow{Elimination} [U \quad \vec{c}] \xrightarrow{Back\ Substitution} [R \quad \vec{x}]$$

$$\vec{b}' = P\vec{b} \qquad L\vec{c} = \vec{b}' \qquad U\vec{x} = \vec{c}$$

**Two steps give us the correct RHS to match U.**

1. **Re-order RHS, if needed:** $\vec{b}' = P\vec{b}$
2. **Solve Lower Triangular system:** $L\vec{c} = \vec{b}'$

- $\vec{c}$ can also be found by simply applying the steps of forward elimination to $\vec{b}$.

**Then we solve with back substitution.**

3. **Solve Upper Triangular system:** $U\vec{x} = \vec{c}$

# Examples

## Finding LU Decomposition

# Find the LU decomposition

- Find L and U such that A=LU, given:

- $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 4 \\ -1 & 0 & 0 & 1 \\ 3 & 3 & 4 & 2 \end{bmatrix}$

# Find the LU decomposition

- **Perform Forward Elimination to get U and encode L:**

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 4 \\ -1 & 0 & 0 & 1 \\ 3 & 3 & 4 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & & 1 & 0 \\ 3 & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1/2 & 1 & 0 \\ 3 & 0 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1/2 & 1 & 0 \\ 3 & 0 & -1 & 1 \end{bmatrix}$$

# Find the LU decomposition

- **Verify:**

$$LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & {}^{1}/_{2} & 1 & 0 \\ 3 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & (2+2) & (2+4) & (2+2) \\ -1 & (-1+1) & (-1+2-1) & (-1+1+1) \\ 3 & 3 & (3+1) & (3-1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 4 & 6 & 4 \\ -1 & 0 & 0 & 1 \\ 3 & 3 & 4 & 2 \end{bmatrix}$$

$= A$

# Find the LU decomposition (with P)

- **Find P, L and U such that PA=LU, given:**

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 2 & 4 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 \end{bmatrix}$$

Question 1: When do we need row changes?

Question 2: How many row changes do we need in this question?

Question 3: How could we find all the needed row changes?

# Find the LU decomposition

- **Perform Forward Elimination to identify row swaps, and generate P:**

$$\begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 2 & 4 & 4 \\ 1 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 2 & 3 \\ 2 & 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & -2 & -2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = P$$

# Find the LU decomposition

- **Perform Forward Elimination to get U and encode L:**

$$PA = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 2 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & -2 & -2 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & & 1 & 0 \\ 1/2 & & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}$$

# Find the LU decomposition

- **Verify:**

$$LU = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 2 & (2-2) & (4-2) & (4-2-2) \\ 1 & 1 & 2 & (2+1) \end{bmatrix} = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 2 & 0 \\ 1 & 1 & 2 & 3 \end{bmatrix} = PA$$

# Example: LU decomposition of a linear system (non square coefficient matrix)

- **Express the given linear system as a pair of triangular systems,** $L\vec{c} = \vec{b}$ **and** $U\vec{x} = \vec{c}$.

$$
\begin{array}{ccccccccc}
2x_1 & + & 3x_2 & + & 4x_3 & + & 4x_4 & = & 2 \\
2x_1 & + & x_2 & + & 4x_3 & + & 3x_4 & = & 5 \\
4x_1 & + & 7x_2 & + & 8x_3 & + & 6x_4 & = & 3
\end{array}
$$

# LU decomposition of a linear system

- **Express in matrix form:**

$$\begin{array}{ccccccccc} 2x_1 & + & 3x_2 & + & 4x_3 & + & 4x_4 & = & 2 \\ 2x_1 & + & x_2 & + & 4x_3 & + & 3x_4 & = & 5 \\ 4x_1 & + & 7x_2 & + & 8x_3 & + & 6x_4 & = & 3 \end{array}$$

$$A\vec{x} = \vec{b}, \qquad A = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 7 & 8 & 6 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

# Find the LU decomposition

- **Perform Forward Elimination on A to get U and encode L**
  **(Do not forget to check whether we need switch rows):**

$$A = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 2 & 1 & 4 & 3 \\ 4 & 7 & 8 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 1 & 0 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}$$

# Find the LU decomposition

- **Define two triangular systems:**

$$A\vec{x} = \vec{b}, A = LU \ \rightarrow L(U\vec{x}) = \vec{b} \rightarrow L\vec{c} = \vec{b}, U\vec{x} = \vec{c}$$

$$L\vec{c} = \vec{b}, \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, \qquad U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

# Examples

## Applications of the LU Decomposition

# Example 1: Solve, given L and U.

- **Solve the linear system $A\vec{x} = \vec{b}$, given:**

$$A = LU,$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

# Solve, given L and U.

- **Set up two triangular systems:**

$$A\vec{x} = \vec{b}, A = LU \rightarrow L(U\vec{x}) = \vec{b} \rightarrow L\vec{c} = \vec{b}, U\vec{x} = \vec{c}$$

$$L\vec{c} = \vec{b}, \qquad L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, \qquad U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -5/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

# Solve, given L and U.

- **Solve** $L\vec{c} = \vec{b}$:

$$L\vec{c} = \vec{b}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/2 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}, \vec{b} = \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & 5 \\ 2 & -1/2 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & -1/2 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1/2 \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} 2 \\ 3 \\ 1/2 \end{bmatrix}$$

# Solve, given L and U

- **Solve** $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -3/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 2 \\ 3 \\ 1/2 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 3 & 4 & 4 & 2 \\ 0 & -2 & 0 & -1 & 3 \\ 0 & 0 & 0 & -5/2 & 1/2 \end{bmatrix} \to \begin{bmatrix} 2 & 3 & 4 & 4 & 2 \\ 0 & -2 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \to \begin{bmatrix} 2 & 3 & 4 & 0 & 14/5 \\ 0 & -2 & 0 & 0 & 14/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix}$$

$$\to \begin{bmatrix} 2 & 3 & 4 & 0 & 14/5 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & 4 & 0 & 7 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 2 & 0 & 7/2 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix}$$

# Solve, given L and U

- **Solve** $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 3 & 4 & 4 \\ 0 & -2 & 0 & -1 \\ 0 & 0 & 0 & -3/2 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 2 \\ 3 \\ 1/2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 7/2 \\ 0 & 1 & 0 & 0 & -7/5 \\ 0 & 0 & 0 & 1 & -1/5 \end{bmatrix} \rightarrow \begin{array}{ccccc} x_1 & + & 2x_3 & = & 7/2 \\ & x_2 & & = & -7/5 \\ & x_4 & & = & -1/5 \end{array}$$

$$\begin{array}{lcl} x_1 & = & 7/2 - 2x_3 \\ x_2 & = & -7/5 \\ x_4 & = & -1/5 \end{array} \rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 7/2 - 2t \\ -7/5 \\ t \\ -1/5 \end{bmatrix}, \forall t \in \mathbb{R}$$

# Example 2: Solve, given P, L and U

- **Solve the linear system $A\vec{x} = \vec{b}$, given:**

$$PA = LU,$$

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix}$$

# Solve, given P, L and U

- **Apply row swaps to RHS:**

$$A\vec{x} = \vec{b}, PA = LU \ \rightarrow L(U\vec{x}) = P\vec{b} \rightarrow \vec{b}' = P\vec{b}, L\vec{c} = \vec{b}', U\vec{x} = \vec{c}$$

$$\vec{b}' = P\vec{b} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

# Solve, given P, L and U

- **Set up two triangular systems:**

$$L\vec{c} = \vec{b}', L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}' = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

# Solve, given P, L and U

- **Solve** $L\vec{c} = \vec{b}'$:

$$L\vec{c} = \vec{b}', L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1/2 & 0 & 0 & 1 \end{bmatrix}, \vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}' = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & -2 & 1 & 0 & -1 \\ 1/2 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & -2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

# Solve, given P, L and U

- **Solve** $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 2 & 4 & 4 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & -2 & -2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 0 & -8 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & -2 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 & 4 & 0 & -8 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 & 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

# Solve, given P, L and U

- **Solve** $U\vec{x} = \vec{c}$:

$$U\vec{x} = \vec{c}, U = \begin{bmatrix} 2 & 2 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -5/2 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -5/2 \\ 2 \end{bmatrix}$$

# Example 3: Solve a system with multiple RHS.

- **Solve the given linear system using Lc=b and Ux=c for two different right-hand sides.**

$$
\begin{array}{rcrcrcrcl}
2x_1 & & & - & 2x_3 & + & 4x_4 & = & d_1 \\
2x_1 & + & 2x_2 & + & 2x_3 & + & 6x_4 & = & d_2 \\
x_1 & - & x_2 & - & 3x_3 & & & = & d_3 \\
 & & 3x_2 & + & 6x_3 & + & 3x_4 & = & d_4
\end{array}
$$

**Case 1:** $(d_1, d_2, d_3, d_4) = (2,2,1,5)$

**Case 2:** $(d_1, d_2, d_3, d_4) = (2,4,2,3)$

# Solve a system with multiple RHS.

- **Express in matrix form:**

$$\begin{array}{rcrcrcrcl} 2x_1 & & & - & 2x_3 & + & 4x_4 & = & d_1 \\ 2x_1 & + & 2x_2 & + & 2x_3 & + & 6x_4 & = & d_2 \\ x_1 & - & x_2 & - & 3x_3 & & & = & d_3 \\ & & 3x_2 & + & 6x_3 & + & 3x_4 & = & d_4 \end{array}$$

**Case 1:** $(d_1, d_2, d_3, d_4) = (2,2,1,5)$

**Case 2:** $(d_1, d_2, d_3, d_4) = (2,4,2,3)$

$$A\vec{x} = \vec{b}, \qquad A = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 2 & 2 & 2 & 6 \\ 1 & -1 & -3 & 0 \\ 0 & 3 & 6 & 3 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 5 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 3 \end{bmatrix}$$

# Solve a system with multiple RHS.

- **Perform Forward Elimination to get U and encode L:**

$$A = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 2 & 2 & 2 & 6 \\ 1 & -1 & -3 & 0 \\ 0 & 3 & 6 & 3 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & -1 & -2 & -2 \\ 0 & 3 & 6 & 3 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & & 1 & 0 \\ 0 & & & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}$$

Note: Do not forget to insert 0 on the fourth row and first column of A!!!!!!!

# Solve a system with multiple RHS.

- **Set up two triangular systems:**

$$A\vec{x} = \vec{b}_i, A = LU \ \rightarrow L(U\vec{x}) = \vec{b}_i \rightarrow L\vec{c}_i = \vec{b}_i, U\vec{x} = \vec{c}_i$$

$$L\vec{c} = \vec{b}, \qquad L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}, \vec{c}_i = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 5 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 3 \end{bmatrix}$$

$$U\vec{x} = \vec{c}, \qquad U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_i = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$$

# Solve a system with multiple RHS.

- **Solve** $L\vec{c}_1 = \vec{b}_1$:

$$L\vec{c}_1 = \vec{b}_1, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}, \vec{c}_1 = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \\ 5 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1/2 & -1/2 & 1 & 0 & 1 \\ 0 & 3/2 & 0 & 1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -1/2 & 1 & 0 & 0 \\ 0 & 3/2 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\vec{c}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

# Solve a system with multiple RHS.

- **Solve** $U\vec{x} = \vec{c}_1$:

$$U\vec{x} = \vec{c}_1, U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_1 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 0 & -2 & 4 & 2 \\ 0 & 2 & 4 & 2 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

- **System is inconsistent, there is no solution for this right-hand side.**

# Solve a system with multiple RHS.

- **Solve** $L\vec{c}_2 = \vec{b}_2$:

$$L\vec{c}_2 = \vec{b}_2, L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1/2 & -1/2 & 1 & 0 \\ 0 & 3/2 & 0 & 1 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 2 \\ 4 \\ 2 \\ 3 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 1 & 1 & 0 & 0 & 4 \\ 1/2 & -1/2 & 1 & 0 & 2 \\ 0 & 3/2 & 0 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & -1/2 & 1 & 0 & 1 \\ 0 & 3/2 & 0 & 1 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\vec{c}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

# Solve a system with multiple RHS.

- **Solve** $U\vec{x} = \vec{c}_2$:

$$U\vec{x} = \vec{c}_2, U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & 0 & -2 & 4 & 2 \\ 0 & 2 & 4 & 2 & 2 \\ 0 & 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & -2 & 4 & 2 \\ 0 & 2 & 4 & 2 & 2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 2 & 0 & -2 & 0 & 10 \\ 0 & 2 & 4 & 0 & 6 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 2 & 0 & -2 & 0 & 10 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -1 & 0 & 5 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

# Solve a system with multiple RHS.

- **Solve** $U\vec{x} = \vec{c}_2$:

$$U\vec{x} = \vec{c}_2, U = \begin{bmatrix} 2 & 0 & -2 & 4 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 & 0 & 5 \\ 0 & 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{array}{ccccc} x_1 & - & x_3 & = & 5 \\ x_2 & + & 2x_3 & = & 3 \\ & x_4 & & = & -2 \end{array}$$

$$\begin{array}{rcl} x_1 & = & 5 + x_3 \\ x_2 & = & 3 - 2x_3 \\ x_4 & = & -2 \end{array} \rightarrow \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 + t \\ 3 - 2t \\ t \\ -2 \end{bmatrix}, \forall t \in \mathbb{R}$$

# Solve a system with multiple RHS.

- **Solve the given linear system using Lc=b and Ux=c for two different right-hand sides.**

$$\begin{array}{rcrcrcrcl} 2x_1 & & & - & 2x_3 & + & 4x_4 & = & d_1 \\ 2x_1 & + & 2x_2 & + & 2x_3 & + & 6x_4 & = & d_2 \\ x_1 & - & x_2 & - & 3x_3 & & & = & d_3 \\ & & 3x_2 & + & 6x_3 & + & 3x_4 & = & d_4 \end{array}$$

$(d_1, d_2, d_3, d_4) = (2,2,1,5)$ → **No solution.**

$(d_1, d_2, d_3, d_4) = (2,4,2,3)$ → $(x_1, x_2, x_3, x_4) = (5 + t, 3 - 2t, t, -2), \forall t \in \mathbb{R}$

# Special Case

## The LDU decomposition.

# LDU decomposition.

- **As derived, L has ones on the diagonal, but U may have any diagonal values.**
- **If we add in only the pivot scaling step of back substitution, U will also have ones at all pivot positions.**

$$A \xrightarrow{Row\ Exchanges} A' \xrightarrow{Elimination} U' \xrightarrow{Pivot\ Scaling} U$$

- **The pivot scaling step can be represented by multiplication by a diagonal matrix.**

# LDU decomposition.

- **The scaling matrix contains the inverse of the diagonal entries:**

$$D^{-1}U' = U$$

$$\begin{bmatrix} 1/u_{11} & 0 & 0 & & 0 \\ 0 & 1/u_{22} & 0 & \cdots & 0 \\ 0 & 0 & 1/u_{33} & & 0 \\ & \vdots & & \ddots & \\ 0 & 0 & & & 1/u_{mm} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} & & u_{1n} \\ 0 & u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & 0 & u_{33} & & u_{3n} \\ & \vdots & & \ddots & \\ 0 & 0 & & & u_{mn} \end{bmatrix} = \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & & u_{1n}/u_{11} \\ 0 & 1 & u_{23}/u_{22} & \cdots & u_{2n}/u_{22} \\ 0 & 0 & 1 & & u_{3n}/u_{33} \\ & \vdots & & \ddots & \\ & & & & \end{bmatrix}$$

In this example: Without loss of generality, we assume that U is an m by n matrix and n >m.

# LDU decomposition.

- **We can also swap the order, using a diagonal matrix to "undo" the scaling.**
- **In this form, we have a diagonal matrix that simply contains the diagonal entries themselves:**

$$DU = U'$$

$$\begin{bmatrix} u_{11} & 0 & 0 & & 0 \\ 0 & u_{22} & 0 & \cdots & 0 \\ 0 & 0 & u_{33} & & 0 \\ & \vdots & & \ddots & \\ 0 & 0 & & & u_{mm} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & & u_{1n}/u_{11} \\ 0 & 1 & u_{23}/u_{22} & \cdots & u_{2n}/u_{22} \\ 0 & 0 & 1 & & u_{3n}/u_{33} \\ & \vdots & & \ddots & \\ 0 & 0 & & & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} & & u_{1n} \\ 0 & u_{22} & u_{23} & \cdots & u_{2n} \\ 0 & 0 & u_{33} & & u_{3n} \\ & \vdots & & \ddots & \\ 0 & 0 & & & u_{mn} \end{bmatrix}$$

In this example: Without loss of generality, we assume that U is an m by n matrix and n >m.

# LDU decomposition.

$$PA = LDU$$

$$A \xrightarrow{\text{Row Exchanges}} A' \xrightarrow{\text{Elimination}} U' \xrightarrow{\text{Pivot Scaling}} U$$

$$A' = PA \qquad A' = LU' \qquad U' = DU$$

- **L and U are triangular (lower and upper) with ones on the diagonal.**
- **A and U are *m*-by-*n* matrices; P, L and D are *m*-by-*m* matrices.**
- **P and L and always non-singular.**
- **D non-singular if (and only if) no zero rows appear during forward elimination; D non-singular if and only if every RHS has a solution.**

# LDU with singular matrices.

- **If $U$ contains zero rows or free variables, $D^{-1}$ often does not exist.**

$$D^{-1}U' = U$$

$$\begin{bmatrix} 1/u_{11} & 0 \\ 0 & 1/0 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & 0 & u_{23} & u_{24} \end{bmatrix} = \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & u_{14}/u_{11} \\ ? & ? & ? & ? \end{bmatrix}$$

$$\begin{bmatrix} 1/u_{11} & 0 & 0 & 0 \\ 0 & 1/u_{22} & 0 & 0 \\ 0 & 0 & 1/u_{33} & 0 \\ 0 & 0 & 0 & 1/0 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} \\ 0 & 1 & u_{23}/u_{22} \\ 0 & 0 & 1 \\ ? & ? & ? \end{bmatrix}$$

# LDU with singular matrices (optional )

- **In some cases with non-square matrices, we can still define $D$, but with ambiguity and/or deviation from the standard.**

$$DU = U'$$

$$\begin{bmatrix} u_{11} & 0 \\ 0 & \mathbf{1} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & u_{14}/u_{11} \\ 0 & \mathbf{0} & u_{23} & u_{24} \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & 0 & u_{23} & u_{24} \end{bmatrix}$$

$$\begin{bmatrix} u_{11} & 0 & 0 & 0 \\ 0 & u_{22} & 0 & 0 \\ 0 & 0 & u_{33} & 0 \\ 0 & 0 & 0 & \mathbf{0} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} \\ 0 & 1 & u_{23}/u_{22} \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \\ 0 & 0 & 0 \end{bmatrix}$$

# LDU with singular matrices (optional).

- **In square matrices, we can always define $D$, with no deviation from the standard.**
- **To achieve this, we don't move zero rows to the bottom; we move them to row(s) that matches the free variable(s)!**

$$DU = U'$$

$$\begin{bmatrix} u_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & u_{33} & 0 \\ 0 & 0 & 0 & u_{44} \end{bmatrix} \begin{bmatrix} 1 & u_{12}/u_{11} & u_{13}/u_{11} & u_{14}/u_{11} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & u_{34}/u_{33} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} & u_{14} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & u_{33} & u_{34} \\ 0 & 0 & 0 & u_{44} \end{bmatrix}$$

# Symmetric Matrix

- Definition: A symmetric matrix has $S^T = S$. This means $S_{ij} = S_{ji}$.
  - The inverse of a symmetric matrix is also symmetric.

$$(S^{-1})^{\mathrm{T}} = (S^{\mathrm{T}})^{-1} = S^{-1}$$

  - The product of $A^T A$ is always symmetric.
  - Best property: If $S = S^T$ is factored into LDU with no row changes, then

$$U = L^T.$$

  - The symmetric factorization of a symmetric matrix is $S = LDL^T$.

# Questions

- Question 1: Is L invertible?
- Answer: Yes, definitely. Because L is a square matrix with 1's on the diagonal.

- Question: When is $A^T A$ invertible?
- Answer: A must have independent columns. Then $Ax = 0$ only if $x = 0$. Otherwise, $Ax = 0\ leads\ to\ \mathrm{A}^T Ax = 0$ which is a contradiction.